

SIO Math Workshop – Mixing Problem Solution

Summary of Known Quantities.

<u>Sample</u>	<u>Water Mass A</u>	<u>Water Mass B</u>
$T = 6\text{ }^{\circ}\text{C}$	$T_A = 4\text{ }^{\circ}\text{C}$	$T_B = 12\text{ }^{\circ}\text{C}$
$S = 35.05\text{ psu}$	$S_A = 34.90\text{ psu}$	$S_B = 35.50\text{ psu}$

Relevant Equations.

Note: The equations below were derived by assuming conservation of volume ($\gamma_B = 1 - \gamma_A$), and substituting into a general conservation of tracer expression ($\gamma_A T_A + \gamma_B T_B = T$).

$$T = \gamma_A T_A + (1 - \gamma_A) T_B \quad \text{Conservation temperature.} \quad (1)$$

$$S = \gamma_A S_A + (1 - \gamma_A) S_B \quad \text{Conservation salt.} \quad (2)$$

Solution.

Note: We have one unknown (γ_A) but two equations—our system is over-constrained! Proceed by solving either (1) or (2) for γ_A . Both choices should yield identical results, barring e.g. sampling errors in our ‘Known Quantities’.

Proceed by solving (1) for γ_A .

$$\begin{aligned} T &= \gamma_A T_A + (1 - \gamma_A) T_B \\ T - T_B &= \gamma_A (T_A - T_B) \\ \gamma_A &= \frac{T - T_B}{T_A - T_B}. \end{aligned}$$

Substitute values from ‘Known Quantities’ table into solution for γ_A .

$$\begin{aligned} \gamma_A &= \frac{T - T_B}{T_A - T_B} \\ &= \frac{6\text{ }^{\circ}\text{C} - 12\text{ }^{\circ}\text{C}}{4\text{ }^{\circ}\text{C} - 12\text{ }^{\circ}\text{C}} \\ \boxed{\gamma_A = 0.75} \quad &\text{Solution is unitless.} \end{aligned}$$

From conservation of volume, we also know:

$$\begin{aligned} \gamma_B &= 1 - \gamma_A \\ &= 1 - 0.75 \\ \boxed{\gamma_B = 0.25}. \end{aligned}$$